Daily Drilling Report
SPD19B-03
3/16/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
6.0	24"	12.00	72.83	31.70	93.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
26" CP, 178.00mKB	0.00	14,627.45	606,913.73		4.60
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
491.00	579.00	569.38	88.00	9.45	9.3

Operations at Report Time
Drilling @ 640m.

Operations Summary
POOH Kick off BHA f/ 27m to surface. RIH Rotary BHA#3. Resume drilling 24" hole f/ 491m to 579m

Operations next 24hrs
Cont' drilling 24" hole.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammad Ali Rezaei	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	3/31/2014
BOP Test		
Chemical Spill Drill		
Fire Drill	3/9/2014	3/11/2014
H2S Drill	3/16/2014	3/31/2014
Heli Crash Drill	2/20/2014	3/14/2014
Medical Drill		
MOB Drill	3/1/2014	3/23/2014
Pit Drill		
Trip Drill		

STOP Cards

Number of Reports: 40

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:30	0.50	24"	D	04	P	POOH kick off BHA#2 f/ 27m to surface. Check the bit (1-1-WT-G-E-I-NO-BHA). *Motor Bearing Gap: 11mm.
00:30	03:00	2.50	24"	D	04	P	M/U Rotary BHA#3 consists of new 24" Insert Bit (Type: ETR14J, IADC: 435,SN: 1250452, Nozzles: 3 x 24 + 1 x 20) + 9 1/8" Shock Sub double box (SN: COR2868AD) + 9 1/2" Float Sub (Float valve inside) + new 23" String Stabilizer (SN: COR2877AD) + 9 1/2" Restrictor Sub (With 32/32" restrictor inside) + 9 1/2" NMDC + 9 1/2" TM Sub (With MWD and Gamma probes inside) and RIH to 21m.
03:00	03:30	0.50	24"	D	12	P	Connect TDS and perform MWD shallow test w/ 600 gpm & 570 psi, ok.
03:30	05:00	1.50	24"	D	04	P	Cont' RIH rest of BHA consists of 9 1/2" NMDC + 17 1/2" COR Stabilizer (COR3130AD) + 1 Std 9 1/2" DC + X/O + 5 Jts 8 1/4" DC + 8" used Cougar drilling jar (COR2985AD) + 2 Jts 8 1/4" DC + X/O + 13 Jts 5 7/8" HWDP to 255m.
05:00	06:30	1.50	24"	D	04	P	RIH Rotary BHA#3 on 5 7/8" DP f/ 255m to 467m. Wash down to 491m.
06:30	14:00	7.50	24"	D	03	P	Resume drilling 24" hole f/ 491m to 545m w/ 850-1000 gpm / 1250-1750 psi / 22-25 kips/ 50-60 rpm & 6 - 19 Kft-lbs. * Drilling with complete loss @ 495m.
14:00	17:00	3.00	24"	D	06	P	After Std down and during back reaming, Driller switch off the pump to change same due to problem in rig pump. While pump on, observe string stuck. Work on it and free same. Have to hard B/R & R/D the Std w/ 750-1000 gpm/ 1200-1800 psi/ 60-100 rpm/ 7-32 Kft-lbs. (String stall continual while B/R & R/D). *** Pump batchs of GG & PHG pills as well. Note: Jar not firing.

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N.I.O.C. Pars Oil & Gas Company

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
17:00	00:00	7.00	24"	D	03	P	Cont' drilling 24" hole f/ 545m to 579m w/ 1000-1100 gpm / 1750-2100 psi / 35 kips/ 50-80 rpm & 7 - 26 Kft-lbs. >>> Sweeping Sequence f/ 491m to 571m: ** At the Middle of each joint: Pump 40 bbl GG. ** At the End of each joint: Pump 20 bbl GG + Circulate 3 min + 40 bbl PHG. ** At the End of Stand: Pump 40 bbl GG + Circulate 5 min + 40 bbl PHG. B/R & R/D the stand. ** Take survey one single off bottom. ** Spot 30 bbl PHG + 20 bbl GG (5 bbl short displacement). Make connection. After connection, first rotate the string then bring up flow rate. >> Drill ahead w/ complete loss, fill up annulus w/ SW. Note: F/ 571m, add 10 bbls to all pill amounts in pill sequences. Mar 17, 2014 00:00 - 06:00 Cont' drilling 24" hole f/ 579m to 640m w/ 1100 gpm / 2100 psi / 35 kips/ 50-40 rpm & 7 - 23 Kft-lbs.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
579.00	Sea Water	8.60				
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
2.00	5.	341	19.0	1,018.0	21.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
DP3	Supply Boat	2/23/2014 23:30		Standby Anchorage.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Bentonite Bulk	4.5	105.0	76.5
Bio-Cide	10.0	139.0	89.0
Caustic Soda	5.0	427.0	261.0
Citric Acid	10.0	210.0	80.0
Guar Gam MP	82.0	1,421.0	1,117.0
Lime	1.0	150.0	66.0
Soda Ash	2.0	151.0	51.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Potable Water	22.0	24.0	87.0
Cement G	0.0	0.0	183.0
Diesel Fuel	11.0	0.0	190.0
Drill Water	110.0	0.0	345.0
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	209.0
Bentonite BB	0.0	0.0	45.0
Bentonite Bulk	5.0	0.0	68.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
2	24	Varel	ETR1GJMRS	1239954	Mill Tooth	115	20/20/20/20
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.22	27.65	173,587.20	183.00	491.00	308.00	11.1	1-1-WT-G-E-I-NO-BHA
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
600	1,000	1,000.0	2,650.0	5	17	69	126
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
37.28	18.37	66.44	33.56	230	221	195	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
491.00	491.00							

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
13 Jts 5 7/8" HWDP	5 7/8	4.250	120.96	259.51
X/O	7	2.810	0.79	138.55
2 Jts 8 1/4" DC	8 1/4	2.810	18.75	137.76
8" Drilling Cougar Jar	8	3.500	5.84	119.01

Daily Drilling Report
SPD19B-03
3/16/2014

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Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
5 Jts 8 1/4" DC	8 1/4	2.810	47.24	113.17
X/O	9 5/8	3.000	1.22	65.93
3 Jts 9 1/2" DC	9 1/2	3.000	28.23	64.71
9 1/2" NMDC	9 1/2	3.100	9.44	36.48
9 1/2" UBHO Sub	9 1/2	3.000	0.83	27.04
9 1/2" Float sub (F.valve inside)	9 1/2		0.77	26.21
9 1/2" Pulser Sub	9 1/2		2.15	25.44
9 1/2" NMDC	9 1/2	4.000	9.46	23.29
9 1/2" Restrictor Sub with restrictor	9 1/2	4.250	0.92	13.83
9 1/2" PNMDC	9 1/2		3.05	12.91
11 1/4" Motor (AKO: 1.83° , Sleeve: 23 1/4")	11 1/4		9.25	9.86
24" Mill Tooth Bit	24		0.61	0.61

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
3	24	Varel	ETR14J	1250452	InserT Bit	435	24/24/24/20
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.60	9.45	33,660.00	491.00	579.00	88.00	9.3	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
850	1,100	1,250.0	2,100.0	22	35	50	80
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
17.16	9.45	100.00	0.00	240	230	219	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
491.00	509.00	1.28	16,000.0	850	50	1,250.0	22	228
509.00	517.00	0.90	18,000.0	1,000	50	1,750.0	25	228
517.00	562.00	5.40	18,000.0	1,000	60	1,750.0	35	228
562.00	575.00	1.15	25,000.0	1,050	80	1,850.0	35	230
575.00	579.00	0.72	23,000.0	1,100	50	2,100.0	35	230

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
13 Jts 5 7/8" HWDP	5 7/8	4.250	120.97	256.31
X/O	7	2.810	0.79	135.34
2 Jts 8 1/4" DC	8 1/4	2.875	18.75	134.55
8" Cougar DA Jar	8	3.500	5.84	115.80
5 Jts 8 1/4" DC	8 1/4	2.875	47.24	109.96
X/O	9 5/8	3.000	1.22	62.72
3 Jts 9 1/2" DC	9 1/2	3.000	28.29	61.50
17 1/2" Stabilizer	17 1/2	3.000	2.47	33.21
9 1/2" NMDC	9 1/2	4.000	9.45	30.74
9 1/2" TM Sub (MWD & Gamma probes inside)	9 1/2	3.125	2.15	21.29
9 1/2" NMDC	9 1/2	4.000	9.45	19.14
9 1/2" Restrictor sub (32/32" restrictor)	9 1/2	4.250	0.91	9.69
23" Stabilizer	23	3.000	2.86	8.78
9 1/2" Float Sub (Float valve insid)e	9 1/2		0.79	5.92
9 1/8" Shock Sub	10 1/8	2.625	4.51	5.13
24" Insert Bit	24		0.62	0.62

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
492.70	231.40	16.20	487.07	-36.50	-40.05	54.08	0.65	0.55
522.00	233.91	17.15	515.14	-41.60	-46.73	62.48	1.22	0.97
550.43	236.10	18.10	542.24	-46.53	-53.79	71.07	1.22	1.00
579.73	240.70	18.30	570.07	-51.32	-61.58	80.14	1.48	0.20
608.69	242.20	18.00	597.59	-55.63	-69.50	89.02	0.58	-0.31

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Asmari		374.50	372.59
Jahrum		493.00	487.36

Daily Drilling Report
SPD19B-03
3/16/2014

General Remarks

Comment

>>> Requirements: (23" Stabilizer) + (Shock Sub) + (200 MT DW) + (PDF MR#1).

>>> Sent:Nil.

>>> Received: Nil.